

The volume of a tea glass

Measured with infinite sums, then checked with water

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About this version. This is an English translation of a school project, with corrections. The idea, the photographs, the fitted curves, the method and the water test are the original ones. Corrected here: one copying error in the volume of the fourth section, which moves the total from 208.6 to 216.9 cm³ (the water test still agrees; Section 5), and a few misprints in intermediate lines that did not affect the numbers. Section 5.1, which says exactly what the water test measures and how precisely, is new.

1 Introduction

Most of my evenings end with a cup of tea, and one evening I looked at the glass I drink it from: a traditional Turkish tea glass (Figure 1). I wanted to find out its volume.



Figure 1. The glass.

2 Aim

To find the volume of the glass using infinite sums. All lengths are in centimetres, areas in square centimetres and volumes in cubic centimetres.

3 The profile of the glass

I put a photograph of the glass into GeoGebra, turned it on its side and scaled it so that one centimetre is one square of the grid (Figure 2). The glass is symmetric about its axis, so it is enough to describe its lower contour and turn it around the axis. The two ends of the rim are

$D(0; 0)$ and $C(0; 6.5)$, so the axis is the line $y = \frac{0+6.5}{2} = 3.25$. The contour changes shape along its length, so I describe it in four sections, each by a parabola.

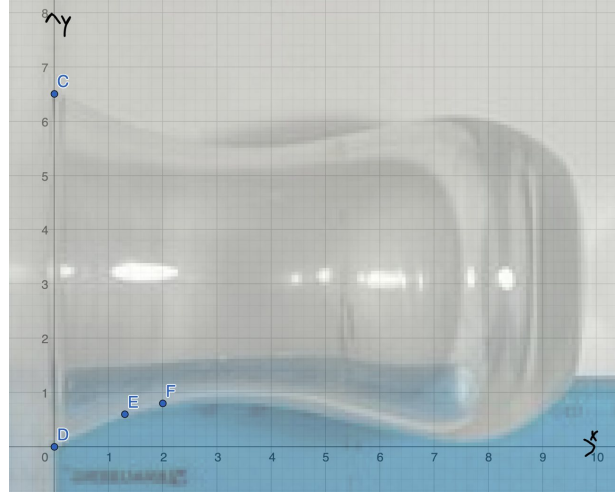


Figure 2. The photograph in GeoGebra, one grid square to a centimetre.

3.1 The first section

The points $D(0; 0)$, $E(1.3; 0.6)$ and $F(2; 0.8)$ look very much like a parabola. A parabola through $(0; 0)$ has the form $y = ax^2 + bx$, so

$$\begin{cases} 0.6 = 1.69a + 1.3b \\ 0.8 = 4a + 2b \end{cases} \implies b = 0.4 - 2a, \quad 0.6 = 1.69a + 0.52 - 2.6a, \quad 0.08 = -0.91a,$$

which gives $a \approx -0.087$ and $b \approx 0.574$: $y = -0.087x^2 + 0.574x$. It follows the contour up to about $x = 3.4$ (Figure 3).

3.2 The second section

After $x = 3.4$ the contour moves away from that parabola. The points $J(4.3; 0.9)$, $H(5.4; 0.7)$ and $G(6; 0.5)$ give

$$\begin{cases} 0.9 = 18.49a + 4.3b + c \\ 0.7 = 29.16a + 5.4b + c \\ 0.5 = 36a + 6b + c \end{cases} \implies \begin{cases} 0.2 = -10.67a - 1.1b \\ 0.2 = -6.84a - 0.6b \end{cases}$$

and, multiplying by 6 and by 11,

$$\begin{cases} 1.2 = -64.02a - 6.6b \\ 2.2 = -75.24a - 6.6b \end{cases}$$

so $1 = -11.22a$, $a \approx -0.089$, then $b \approx 0.681$ and $c \approx -0.382$: $y = -0.089x^2 + 0.681x - 0.382$, which follows the contour up to about $x = 6$ (Figure 4).

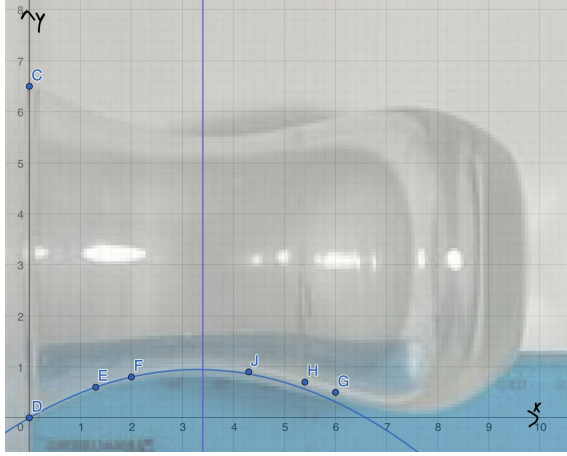


Figure 3. The first parabola.

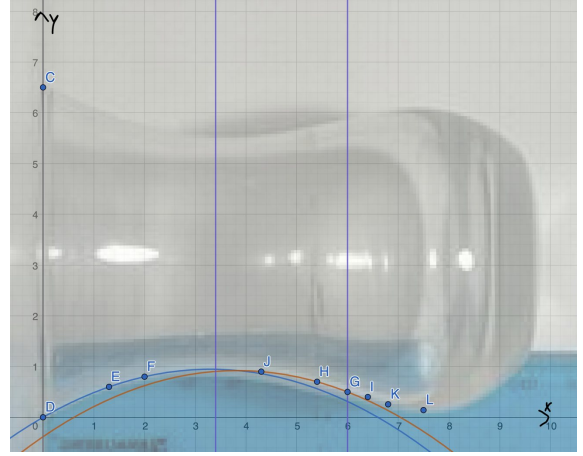


Figure 4. The first two parabolas.

3.3 The third section

The points $I(6.4; 0.4)$, $K(6.8; 0.25)$ and $L(7.5; 0.1)$ give, after subtracting the equations in pairs and multiplying by 7 and by 4,

$$\begin{cases} 0.15 = -5.28a - 0.4b \\ 0.15 = -10.01a - 0.7b \end{cases} \Rightarrow \begin{cases} 1.05 = -36.96a - 2.8b \\ 0.6 = -40.04a - 2.8b \end{cases}$$

so $0.45 = 3.08a$, $a \approx 0.146$, then $b \approx -2.301$ and $c \approx 9.146$: $y = 0.146x^2 - 2.301x + 9.146$, up to about $x = 7.5$ (Figure 5).

3.4 The fourth section

The points $R(7.9; 0.1)$, $Q(8.8; 0.5)$ and $N(9.2; 0.8)$ give, multiplying by 4 and by 9,

$$\begin{cases} -0.4 = -15.03a - 0.9b \\ -0.3 = -7.2a - 0.4b \end{cases} \Rightarrow \begin{cases} -1.6 = -60.12a - 3.6b \\ -2.7 = -64.8a - 3.6b \end{cases}$$

so $1.1 = 4.68a$, $a \approx 0.235$, then $b \approx -3.48$ and $c \approx 12.926$: $y = 0.235x^2 - 3.48x + 12.926$. The glass ends at $x = 9.6$ (Figure 6).

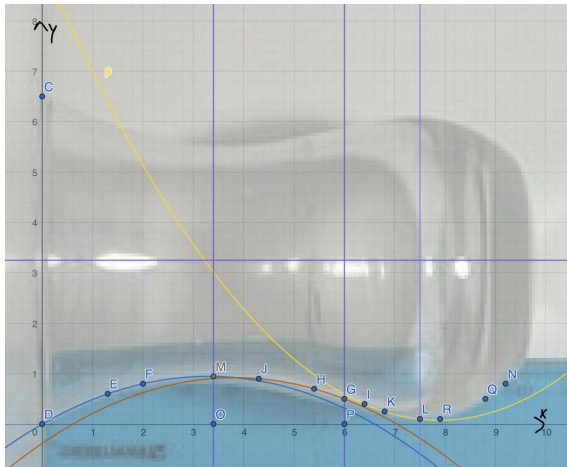


Figure 5. The third parabola added.

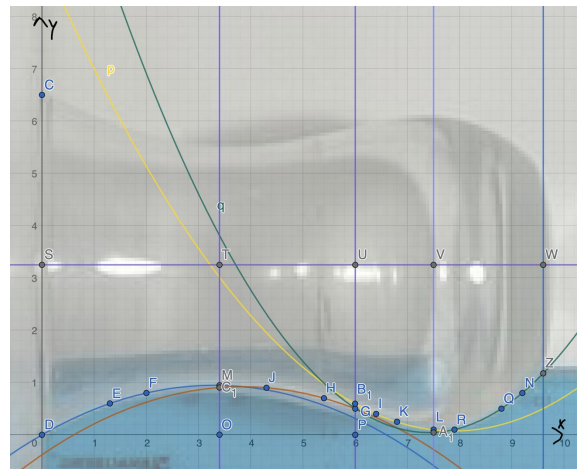


Figure 6. All four sections, and the axis $y = 3.25$.

Together, the lower contour of the glass is

$$f(x) = \begin{cases} -0.087x^2 + 0.574x, & 0 \leq x \leq 3.4, \\ -0.089x^2 + 0.681x - 0.382, & 3.4 < x \leq 6, \\ 0.146x^2 - 2.301x + 9.146, & 6 < x \leq 7.5, \\ 0.235x^2 - 3.48x + 12.926, & 7.5 < x \leq 9.6. \end{cases}$$

4 The volume, by discs

The method. Take one section, from x_0 to $x_0 + L$, and cut it into n slices of equal width $\frac{L}{n}$, where $n \rightarrow \infty$. Turned around the axis, each slice is a thin disc. The i -th one sits at $x_i = x_0 + i\frac{L}{n}$ and has radius

$$r_i = 3.25 - f(x_i),$$

so its volume is $\pi r_i^2 \cdot \frac{L}{n}$, and the volume of the section is

$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n \pi r_i^2 \frac{L}{n}.$$

Writing $u = x - x_0$ for the distance into the section, r^2 is a polynomial of degree four in u , say $r^2 = A_4u^4 + A_3u^3 + A_2u^2 + A_1u + A_0$, and $u_i = i\frac{L}{n}$. So the sum needs the well-known formulas

$$\begin{aligned} \sum_{i=1}^n i &= \frac{n(n+1)}{2}, & \sum_{i=1}^n i^2 &= \frac{n(n+1)(2n+1)}{6}, \\ \sum_{i=1}^n i^3 &= \left(\frac{n(n+1)}{2}\right)^2, & \sum_{i=1}^n i^4 &= \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}. \end{aligned}$$

For the fourth powers, for example,

$$\left(\frac{L}{n}\right)^5 \sum_{i=1}^n i^4 = L^5 \left(\frac{1}{5} + \frac{1}{2n} + \frac{1}{3n^2} - \frac{1}{30n^4}\right) \rightarrow \frac{L^5}{5},$$

because the terms with n in the denominator vanish as $n \rightarrow \infty$. In the same way the other powers give $\frac{L^4}{4}$, $\frac{L^3}{3}$ and $\frac{L^2}{2}$, and therefore

$$V = \pi \left(A_4 \frac{L^5}{5} + A_3 \frac{L^4}{4} + A_2 \frac{L^3}{3} + A_1 \frac{L^2}{2} + A_0 L \right).$$

4.1 The first section, $0 \leq x \leq 3.4$

Here $x_0 = 0$, so $u = x$, $L = 3.4$ and

$$\begin{aligned} r &= 3.25 + 0.087u^2 - 0.574u, \\ r^2 &= 0.007569u^4 - 0.099876u^3 + 0.894976u^2 - 3.731u + 10.5625. \end{aligned}$$

$$\begin{aligned} V_1 &= \pi \left(0.007569 \cdot \frac{3.4^5}{5} - 0.099876 \cdot \frac{3.4^4}{4} + 0.894976 \cdot \frac{3.4^3}{3} - 3.731 \cdot \frac{3.4^2}{2} + 10.5625 \cdot 3.4 \right) \\ &= \pi (0.688 - 3.337 + 11.725 - 21.565 + 35.913) \approx 73.59. \end{aligned}$$

4.2 The second section, $3.4 < x \leq 6$

Here $x = 3.4 + u$ and $L = 2.6$. The radius is $r = 3.632 + 0.089x^2 - 0.681x$, which in u is

$$\begin{aligned} r &= 2.3454 - 0.0758u + 0.089u^2, \\ r^2 &= 0.0079u^4 - 0.0135u^3 + 0.4232u^2 - 0.3556u + 5.5011. \end{aligned}$$

$$\begin{aligned} V_2 &= \pi(0.0079 \cdot \frac{2.6^5}{5} - 0.0135 \cdot \frac{2.6^4}{4} + 0.4232 \cdot \frac{2.6^3}{3} - 0.3556 \cdot \frac{2.6^2}{2} + 5.5011 \cdot 2.6) \\ &= \pi(0.188 - 0.154 + 2.480 - 1.202 + 14.303) \approx 49.06. \end{aligned}$$

4.3 The third section, $6 < x \leq 7.5$

Here $x = 6 + u$ and $L = 1.5$. The radius is $r = -5.896 - 0.146x^2 + 2.301x$, which in u is

$$\begin{aligned} r &= 2.654 + 0.549u - 0.146u^2, \\ r^2 &= 0.0213u^4 - 0.1603u^3 - 0.4736u^2 + 2.9141u + 7.0437. \end{aligned}$$

$$\begin{aligned} V_3 &= \pi(0.0213 \cdot \frac{1.5^5}{5} - 0.1603 \cdot \frac{1.5^4}{4} - 0.4736 \cdot \frac{1.5^3}{3} + 2.9141 \cdot \frac{1.5^2}{2} + 7.0437 \cdot 1.5) \\ &= \pi(0.032 - 0.203 - 0.533 + 3.278 + 10.566) \approx 41.28. \end{aligned}$$

4.4 The fourth section, $7.5 < x \leq 9.6$

Here $x = 7.5 + u$ and $L = 2.1$. The radius is $r = -9.676 - 0.235x^2 + 3.48x$, which in u is

$$\begin{aligned} r &= 3.2053 - 0.045u - 0.235u^2, \\ r^2 &= 0.0552u^4 + 0.0212u^3 - 1.5044u^2 - 0.2885u + 10.2736. \end{aligned}$$

$$\begin{aligned} V_4 &= \pi(0.0552 \cdot \frac{2.1^5}{5} + 0.0212 \cdot \frac{2.1^4}{4} - 1.5044 \cdot \frac{2.1^3}{3} - 0.2885 \cdot \frac{2.1^2}{2} + 10.2736 \cdot 2.1) \\ &= \pi(0.451 + 0.103 - 4.644 - 0.636 + 21.575) \approx 52.93. \end{aligned}$$

Correction. In the original, the fourth term of this line was copied from the third section (3.278, which is $2.9141 \cdot \frac{1.5^2}{2}$). The correct value is $0.2885 \cdot \frac{2.1^2}{2} = 0.636$, which raises V_4 from 44.67 to 52.93.

4.5 The whole glass

$$V = V_1 + V_2 + V_3 + V_4 \approx 73.59 + 49.06 + 41.28 + 52.93 \approx 216.9 \text{ cm}^3.$$

5 Checking the result

To make sure the calculation was right, I did an experiment. I put the glass into a measuring beaker of water and made sure it was completely under water (Figure 7); the level was at 825 ml. Then I lifted the glass out carefully, full of water (Figure 8). The level dropped to just below 625 ml, but clearly above 600 ml (Figure 9). So the body I lifted out had a volume of a little over 200 ml and clearly under 225 ml. The calculated 216.9 cm^3 (one cubic centimetre is one millilitre) is inside that range.

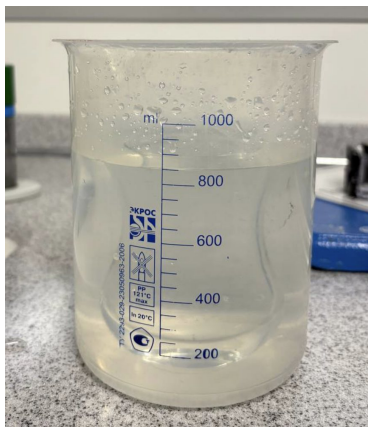


Figure 7. The glass under water: 825 ml.



Figure 8. Lifted out, full of water.

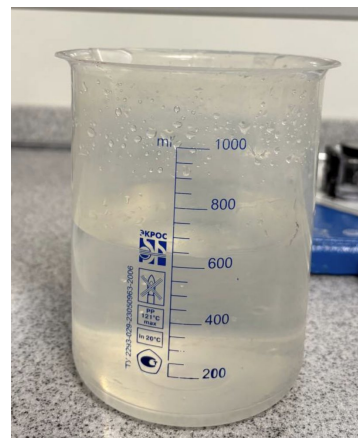


Figure 9. Afterwards: just below 625 ml.

5.1 What the test measures, and how precisely

Lifting the glass out *full of water* takes away the glass itself together with the water inside it: everything enclosed by the outer surface of the glass, up to the rim. That is exactly what turning the outer contour around the axis computes, so the calculation and the test measure the same thing. (The capacity of the glass, how much tea it holds, is smaller by the volume of the glass itself.)

The test is as precise as the beaker. The level was read between graduation marks, so it places the volume between 200 and 225 ml, and the calculation lands inside that window rather than on a single number. A kitchen scale would narrow it to a gram, since a millilitre of water weighs one gram. The calculation has its own limits too: the contour points were read by eye from a photograph, which a camera slightly distorts, and each parabola passes through only three of them, so neighbouring parabolas miss each other by up to a millimetre at the joins (at $x = 6$ they give 0.50 and 0.60).

6 Reflection

This project suggests others. One is to measure the inner volume of the glass; with both volumes and the mass of the glass, it would be possible to find the density of the glass itself, by dividing the mass by the difference of the two volumes. The same way of measuring volume could be used in construction: comparing the volume of a building drawn on a computer with the volume of the building actually built would say something about the quality of the work.

I enjoyed this project, although not everything went as planned. At first I wanted to measure a plush Minion toy. But the toy changed shape at the slightest touch and was hard to put back the way it was, so it was not symmetric at all, and it had many separate parts besides its body (goggles, arms and legs). Measuring it would have taken far too long.